IBM Z Datathon 2026

Student Questions – Technical FAQ & IBM Response

As a follow-up to the IBM SME-led engagement sessions, we have consolidated the substantive questions raised by students across the **IBM Z & LinuxONE**, **AI on IBM Z**, and **Security & Identity** sessions and IBMs response to these questions.

A. IBM Z & LinuxONE – Fundamentals

1. Is LinuxONE a new Linux distribution?

IBM LinuxONE is **not a Linux distribution**. It is an enterprise computing platform designed specifically to run Linux workloads.

You can run familiar Linux distributions like Red Hat, Ubuntu and Suse and use many of the same open-source tools, programming languages, containers, databases and frameworks you may already know. The important difference is the underlying IBM Z architecture and the enterprise capabilities it provides for scalability, security, reliability and workload consolidation.

Think of LinuxONE as a powerful enterprise Linux server platform—not a new version of Linux.

2. What is reliability in the context of IBM Z?

Reliability means designing a computing system to **continue operating correctly and consistently, even when individual components experience problems**.

IBM Z systems include extensive redundancy, error detection, fault isolation and recovery capabilities. Components can often be repaired or replaced without stopping the applications running on the system.

This matters for workloads such as banking, payments, healthcare and other services where downtime can have a major impact.

IBM LinuxONE 5, for example, is designed for availability of up to **99.999999%** in appropriately configured environments.

3. Where do you see IBM Z and LinuxONE technology being used in the future?

IBM Z and LinuxONE are increasingly relevant wherever organizations need to combine **AI, large-scale transaction processing, security, data and hybrid cloud**.

Examples include financial services, healthcare, government, retail, transportation and other industries processing large volumes of important or sensitive data.

One particularly interesting direction is **AI close to the transaction and the data**. Instead of sending data to a separate system, organizations can perform AI inference on the same platform where important applications and data already reside.

Examples could include detecting fraud during a payment, identifying suspicious activity, assessing transaction risk or making real-time recommendations.

4. Why should organizations choose IBM Z over other software or platforms?

IBM Z isn't necessarily the right platform for every workload. Organizations typically consider it when they need combinations of:

- Very high availability and resiliency
- High-volume transaction processing
- Strong security and encryption
- Large-scale workload consolidation
- Predictable performance
- AI integrated with enterprise transactions and data
- Efficient use of data-center space and energy

IBM Z is also designed to participate in **hybrid-cloud architectures**, so choosing IBM Z doesn't mean that everything an organization operates has to run on IBM Z.

5. What are cloud x86 servers?

Most cloud servers are based on the **x86 processor architecture**, commonly using processors from Intel or AMD.

When you create a virtual machine with a public cloud provider, it will often be running on an x86 physical server somewhere in that provider's data center.

IBM Z and LinuxONE use a different processor architecture called **IBM Z architecture (s390x)**.

Linux applications can run on both architectures, although software sometimes needs to be compiled or packaged specifically for the architecture on which it will run.

6. What are LPARs?

LPAR stands for **Logical Partition**.

An LPAR allows one physical IBM Z or LinuxONE system to be divided into multiple isolated computing environments. Each partition can be assigned its own processors, memory and other resources and can run independently.

An easy analogy is dividing a large building into secure apartments. Everyone shares the same physical building, but each apartment operates as its own private space.

LPAR technology is implemented at a very low level of the IBM Z platform and provides strong isolation between workloads.

7. What are modular systems?

A modular system is designed so that computing resources can be configured and expanded using components rather than requiring the entire environment to be replaced.

IBM Z systems can be configured with different amounts of processing capacity, memory, I/O and other resources according to an organization's requirements.

This allows organizations to start with the capacity appropriate for their workload and expand or modify the environment as their requirements change.

B. IBM Z – Performance & Efficiency

9. Is IBM Z faster than traditional server clusters?

It depends on the workload.

It isn't accurate to say that IBM Z is simply "faster than x86" for every application. Different architectures are optimized for different workloads.

IBM Z is particularly strong at **high-volume enterprise transaction processing, large-scale data processing, workload consolidation and applications requiring consistent performance and very high availability**.

Another advantage is that processing, databases and AI inference can potentially occur on the same system. That can eliminate some of the network communication that would otherwise occur between separate servers.

For the Shooting Stars ZDatathon, this is an important concept: **performance isn't only CPU speed. Architecture and where the data and processing occur matter too.**

10. How efficient is IBM Z in terms of energy consumption?

IBM Z and LinuxONE are designed to consolidate very large numbers of workloads onto relatively few physical systems.

That can significantly reduce the number of servers, network connections, floor space and power required for certain large-scale workloads.

For example, IBM reports that for a specific cloud-native containerized workload comparison, LinuxONE Emperor 5 can use **up to 65% less power** than the compared x86 solution.

The exact savings depend heavily on the workload and configuration, so this shouldn't be interpreted as meaning every application automatically uses 65% less power.

11. What is clock speed, and how does it relate to IBM Z performance?

Clock speed, usually measured in GHz, indicates how many processor clock cycles occur each second.

But **GHz alone doesn't tell you how fast a computer will run an application**.

Processor architecture, cache, memory, I/O, instruction efficiency, number of processors, specialized accelerators and software optimization all affect performance.

IBM Z is a good example of why comparing computers based only on GHz can be misleading. It is designed as an integrated system for handling large numbers of enterprise workloads and transactions, so overall system architecture matters much more than one processor specification.

C. IBM Z & Applications

D. IBM Z – Learning & Careers

13. How can IBM Z and LinuxONE technology help beginners improve their skills and build better career opportunities?

You don't have to begin by learning traditional mainframe programming.

If you already know **Linux, Python, Java, JavaScript, Git, containers, Kubernetes, databases, APIs or AI/ML**, many of those skills can be applied to IBM Z and LinuxONE.

The Datathon is a great example. Participants can take technologies they already understand and learn how those technologies operate on enterprise infrastructure.

IBM also provides **IBM Z Xplore**, where students can complete hands-on challenges, learn IBM Z technologies and earn digital credentials.

Knowledge of both modern cloud/open-source technologies and enterprise computing can create an interesting combination of skills for a career.

E. AI on IBM Z – Datathon & Technical Requirements

14. What platforms or technology stacks can participants use for their Datathon projects?

Participants should feel free to use modern technologies appropriate for their solution.

Depending on the project, that might include:

Python, Java, JavaScript/Node.js, Linux, databases, APIs, containers, open-source AI/ML frameworks and other appropriate development tools.

The goal isn't to force every team into the same technology stack. The important part is developing an innovative AI/data solution and demonstrating meaningful use of IBM Z/LinuxONE.

15. Is using IBM Z mandatory for every project?

The intention of the IBM Z Datathon is for projects to demonstrate meaningful use of **IBM Z/LinuxONE technology**.

That doesn't mean every component of your solution has to run there. A project might include a web or mobile frontend, external APIs or other cloud services while using IBM Z/LinuxONE for an important part of the backend, data processing, AI or application architecture.

The key is that IBM Z/LinuxONE should be a meaningful part of the solution rather than simply being mentioned in the presentation.

16. Are participants expected to use IBM Z technologies directly, or can IBM Z be integrated with other technologies?

For this IBM Z Datathon we do expect that participants directly use the LinuxONE platform for their submissions. You can combine IBM Z/LinuxONE with other technologies as well but I it is important that you directly use the LinuxONE Platform.

17. Are there any specific requirements for where participants should deploy their prototype?

Participants will have access to the **LinuxONE Community Cloud** for development and deployment.

Projects should use the provided IBM Z/LinuxONE resources in accordance with the Datathon instructions.

18. What IBM Z technologies are mandatory for the Datathon?

The use of the **LinuxONE Platform through the LinuxONE Community Cloud is** mandatory however, you have the freedom to use whatever technology you choose on the platform itself.

That means participants can build with familiar Linux and open-source technologies while still gaining experience developing on IBM Z architecture.

19. How can beginners effectively use IBM Z technologies in their Datathon projects?

Start simple.

Don't spend the entire 24-hour Datathon trying to learn every aspect of IBM Z.

If you know Python, for example, start by getting your Python application running on your LinuxONE Community Cloud virtual server. Then add your data processing, database, API or AI components.

A simple architecture that actually works is much more valuable than an extremely complicated architecture that never gets finished.

Mentors and IBM resources can help you explore more advanced capabilities as your project develops.

20. What hardware and software requirements should participants have ready and enabled before the Datathon?

Participants should have a reliable computer and internet connection and should verify that they can access the accounts and resources provided for the Datathon.

It is helpful to have:

- A laptop or desktop computer
- Reliable internet access
- A modern web browser
- An SSH client
- A code editor or IDE such as VS Code
- Git/GitHub or another source-control tool
- Any local development tools required by your team's project
- Your Datathon and IBM/LinuxONE access credentials

Most importantly, **test your access before the competition starts**. A 24-hour Datathon is not the time to discover that you cannot SSH into your development environment.

F. AI & IBM Z

21. Can training AI models be made more efficient using IBM Z?

Potentially, depending on the model and workload, but it is important to distinguish **training from inference**.

Training creates or updates a model and can require enormous computational resources. Very large generative-AI models are commonly trained using large GPU or accelerator clusters.

Inference happens after training: you give the trained model new data and ask it to make a prediction, classification or other output.

A major strength of current IBM Z and LinuxONE systems is **AI inference close to enterprise applications and data**. IBM z17 and LinuxONE 5 include Telum II processors with integrated AI acceleration, and additional AI acceleration is available through IBM Spyre Accelerator technology.

For example, an AI model could be trained elsewhere and then deployed near transaction data on IBM Z to evaluate transactions in real time.

For Datathon participants, both approaches are possible: **you don't necessarily have to train your model on IBM Z to make meaningful use of AI on IBM Z.**

G. Data & AI Security

23. How does IBM address AI vulnerabilities related to fetching and storing data?

AI security should be approached in layers.

Organizations need to consider:

Data in transit – protecting information while it moves between systems.

Data at rest – protecting information stored in files, databases and storage.

Data in use – protecting sensitive information while applications are processing it.

IBM Z and LinuxONE provide hardware- and software-based security capabilities including encryption, protected key management, access controls, workload isolation and confidential-computing technologies.

But infrastructure security alone isn't enough. Developers still need secure authentication, authorization, API security, proper secrets management, software updates and careful control over what data an AI application is allowed to access.

H. AI Coding Tools

I. Careers & IBM Z

25. How can future backend engineers contribute to IBM and IBM Z Systems?

There are many opportunities beyond traditional mainframe programming.

Modern IBM Z environments need engineers who understand:

- Java
- Python
- C/C++
- APIs and REST services
- Linux
- Containers and Kubernetes/OpenShift
- Databases
- DevOps and CI/CD
- Cloud architecture
- Cybersecurity
- AI and machine learning
- Observability and automation

There are also careers involving traditional and modern z/OS technologies such as COBOL, JCL, CICS, IMS, Db2, system programming and system administration.

One particularly valuable skill is being able to **connect modern applications and APIs with enterprise systems and data**.

26. What types of jobs and skill sets is IBM currently looking for?

IBM hires across many different areas, and specific openings continually change.

Relevant career areas can include software engineering, AI and machine learning, data science, cybersecurity, cloud and infrastructure engineering, DevOps, consulting, research, technical sales and IBM Z systems engineering.

For students, a strong foundation in **programming, Linux, Git, APIs, data, security and problem solving** is valuable.

Don't feel that you have to learn every technology. Develop strong fundamentals, become very good at a few areas, and demonstrate that you can continue learning.

A working Datathon project that you can explain clearly can itself become a useful portfolio example.

J. Mentorship & Participant Support

27. Will IBM provide mentorship to teams during the Datathon?

Yes. The Datathon is designed as a hands-on learning experience, and the event includes **expert mentorship** to help participants as they develop their solutions.

Mentors can help participants understand technologies, troubleshoot problems and think through possible architectures and approaches.

Teams should still make their own design decisions and build their own solutions—the mentors are there to help you learn and move forward when you get stuck.

K. Security & Identity

28. What are encrypted keys?

An encrypted key is a cryptographic key that has itself been encrypted—usually so that it can be stored or transported more securely.

29. If a person wants to log in to their account from another device, how does the authentication/key mechanism work?

After a user registers for their account on the IBM LinuxONE Community Cloud they will log into a dashboard that allows them to provision a linux server on LinuxONE. At that time a user can either create a public/private key pair, in which the private key is downloaded to the user's workstation and the private key is stored on the linux server. Or if the user already has a

public/private key pair that they use for other systems, they can upload the public key to the linux server during creation of the server and use their private key to log in via SSH.

31. Which programming language should I learn to provide security for my software?

There isn't one "security programming language."

Security is a property of how software is **designed, implemented, configured and operated**.

For Datathon participants, languages such as **Python, Java, JavaScript/TypeScript, Go, C/C++ and Rust** can all be relevant depending on what you want to build.

Rather than choosing a language solely for security, learn secure-development concepts such as:

authentication, authorization, encryption, TLS, secrets management, input validation, API security, dependency management and secure coding practices.

Those concepts transfer between programming languages.

L. Certificates (Vaidehi we can discuss this on our call)

33. When will participants receive their participation certificates?

Bruce – I don't' know that answer to this related to the Z Datathon